

Retro Tron Battle — Hardware Test Checklist

Classic Computing 2026 — Celle, 10.–11. Oktober 2026

0. Before touching hardware

- [] Confirm server LAN IP for today. All four clients hardcode 192.168.17.158 (HOST\$/ho\$/HO\$) — edit all four .bas files if it differs, re-tokenize/transfer.
- [] Start server: `python3 server/tron_server.py` — confirm startup banner shows TCP 6502 / HTTP 8080 (or whatever the clients expect).
- [] Confirm build artifacts are today's: `clients/c64/client.prg`, `clients/c64/wic64.prg`, `disk-images/localtest.d64` (currently untracked in git) — not stale leftovers.
- [] If testing the photo feature: `sudo python -m pyftplib -p 21` from `assets/test_ftproot/`, PIN MUSTER (or 0001 / 0002 / 4711 / 0815).

1. Joystick test first — every platform, before the game client

Run the standalone diagnostic before touching the actual game client. No server, no networking — just confirms the stick and its port are wired correctly. Once a stick checks out, physically label it (e.g. a tape strip: "ATARI — PORT 1") with which platform/port it was tested and connected to. With three machines and swappable sticks, a mixed-up cable is an easy way to lose an hour chasing a phantom bug.

- [] Atari: `tools/atari_joystick_test.bas` — check both STICK(0) and STICK(1), label the stick.
- [] C64: `tools/c64_joystick_test.bas` — confirm PORT 2 (\$dc00) responds and PORT 1 (\$dc01) stays idle while only touching the stick in port 2. If port 1 also twitches, that's the keyboard-scan ghosting bug from CLAUDE.md resurfacing — stop and re-check wiring before moving on. Label the stick.
- [] CPC: `tools/cpc_joystick_test.bas` — check JOY(0) responds in all 4 directions, label the stick.

2. Per-platform client status — what's actually unverified

Platform	Last verified on hardware	Watch for
Atari / FujiNet	Stable, but before the 2026-09-12 UI simplification	Full match incl. restart — client must use RUN, not GOTO, between matches
C64 Meatloaf	Session-mismatch fix from 2026-09-12, not reverified	Join succeeds despite lowercase-mangling; joystick physically in PORT 2
C64 WiC64	Never tested on real hardware, ever	See section 3 below — today is the first test
CPC / M4	Stable pre-simplification, fix applied preventively, not reverified	HTTPMEM blocking joystick during calls is expected, not a bug

3. C64 WiC64 — special attention (first real-hardware test)

- [] Disk has FOTOFIX.C000 alongside the client (`clients/c64/wic64-driver/`).
- [] Boot: lowercase load"fotofix.c000" actually finds the file.
- [] /join and /tick URLs are lowercase now — confirm no "ERR UNKNOWN SESSION".
- [] Joystick on PORT 2 with the new CLI/RTS stub (line 46, poke828,88:poke829,96) actually fixes the ghosting.
- [] Watch the temporary debug print (lines 585–588: raw joystick value + direction) — remove once confirmed working, before the event.
- [] Open question: does SYS49152 hold up under rapid repeated /tick polling, vs. the driver's original rare-call design?

4. Cross-cutting checks (single platform)

- [] Play at least 2 consecutive matches per platform, not just one — this project's worst bugs were “works once, fails the second time” (Atari GOTO-vs-RUN, C64 device-path scrambling).
- [] Server display: logos load per match, scroll text readable at SCROLL_SPEED=2 (just reduced from 4 recently).
- [] Roughly time how long it feels between moving the stick and the on-screen character turning, per platform — this gives real latency numbers for section 5, instead of guessing.

5. Smoke test: all clients battle each other

The server only ever pairs two players at a time (first two connections) — plan sequential 1v1 matches, not a 4-way free-for-all.

Suggested pairing order:

- [] Atari vs C64 (Meatloaf) — both baseline-stable; confirms cross-play/pairing/win-detection itself works before anything experimental joins in.
- [] Atari vs C64 (WiC64) — fastest client against the least-tested one.
- [] C64 (Meatloaf) vs CPC — two HTTP-polling clients against each other.
- [] C64 (WiC64) vs CPC — the two least-proven clients against each other.
- [] Atari vs CPC.
- [] Optional, only with a second physical C64: Meatloaf vs WiC64 directly.

On tick speed: server/tron_server.py runs one fixed-rate game loop (TICK_RATE = 8/sec, line 232) — movement itself advances at the same rate for both players no matter the platform, so that part is already fair. What actually differs per platform is how fast a direction change *reaches* the server: Atari keeps a raw TCP socket open and sends MOVE the instant you steer; the HTTP-polling clients (Meatloaf, WiC64, CPC) can only submit a new direction once their current blocking /tick round-trip completes. If that round-trip takes longer than 125 ms (one server tick) on real WiFi hardware, that platform's steering will be coarser than Atari's — that's an inherent consequence of the transport, not a bug to chase.

Use the rough latency numbers from section 4 to see if the gap is actually severe. Only reconsider TICK_RATE if it clearly is — don't tune it blindly the night before the event.

If a pairing misbehaves and you need to isolate whether it's the pairing or one specific client, bots/tron_bot.sh can stand in for the TCP side (Atari-protocol) as a predictable opponent.

6. Suggested order for the day

Server smoke test alone → joystick tests for all three platforms, with labeling (section 1) → solo client test per platform: **C64 Meatloaf** → **Atari** → **CPC** → **C64 WiC64 last** (most experimental — if it doesn't pan out today, Meatloaf is the fallback for CC2026, so that decision should get made with time still on the clock) → once all four are solo-verified, run the cross-platform smoke test pairings from section 5.